Strict Cross-Slice Order and Line Batching in the Exact Fisher Network

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Research release v0.2.0
19 August 2026

Abstract

At one step of the Mehta–Patel exact Fisher network, a daughter allocation x fixes the next table column. Write $B(x) = A(x)L(R - x; C')$ for the largest complete branch weight below it. Network-induced M-concavity of $\log L$, strengthened by the strict first-column multinomial factor, yields a strict cross-slice rhombus. A within-line ordering lemma then implies monotone movement of leftmost daughter modes across adjacent three-coordinate slices and an amortized $O(m + s)$ peak-query sweep. A controlled native ablation confirms the query reduction while finding wall-clock timing essentially neutral. The accompanying three-row implementation also evaluates daughter-line batching, exact outer-interval acceptance, interval-mass aggregation and prefix reuse. These mechanisms reduce work against matched unbatched controls on prefix-heavy tables; retained difficult workloads remain slower than base R FEXACT. The release keeps exact-rational verification and FEXACT-compatible floating computation as explicit numerical scopes.

1 Introduction

The Mehta–Patel network computes the Fisher–Freeman–Halton exact tail for a contingency table with fixed margins by merging equal residual states and trimming complete subtrees with exact continuation bounds [3, 4]. FEXACT adds packed state representations, shared bound caches and ordered prefix storage [2]. Later work develops exact path trimming, discrete-convex transportation formulations and one-dimensional daughter ordering [1, 8, 7].

This paper isolates one structural result within a broader implementation. Sections 2–6 prove a strict cross-slice branch inequality and its monotone-leftmost-mode consequence. Section 7 states exactly what that consequence changes in an implementation. Section 8 reports controlled ablations and the separate effects of line batching and accept-only pruning. The theorem controls peak location; interval acceptance, arc reduction, state reduction and memory behavior are measured separately.

The conditional exact test and network construction are due to Mehta and Patel [3, 4]; the practical FEXACT refinements are due to Clarkson, Fan and Joe [2]; the network-induced discrete-convex structure used in the proof is due to Murota [5, 6]; and the later path-trimming, transportation and daughter-ordering developments are due respectively to Aoki; Suzuki, Aoki and Murota; and Requena and Martín Ciudad [1, 8, 7]. This paper establishes the strict cross-slice rhombus, its monotone-leftmost-mode and amortized peak-query consequences, and a controlled implementation study. The cited literature supplies the exact test, network algorithm, FEXACT, network induction, transportation extrema and one-dimensional daughter ordering.

In standard array language, the cross-slice result is a strict multiplicative Monge-type inequality, equivalently strict log-submodularity, on each feasible elementary slice rhombus.

2 Exact Fisher network and branch bounds

Let $Y = (y_{ab})$ be a nonnegative $r \times c$ integer table with fixed row and column margins. Conditional probability is proportional to

$$\frac{1}{\prod_{a,b} y_{ab}!}.$$

The exact two-sided tail used here contains every table no more probable than the observed table. At a residual network state, let $R \in \mathbb{Z}_+^r$ be the row margins, let the next column total be m, and let the later column margins be $C' = (c_1, \ldots, c_q)$. For a feasible daughter allocation x, define

$$A(x) = \binom{m}{x_1, \ldots, x_r}, \qquad L(y; C') = \max_Z \prod_{b=1}^{q} \binom{c_b}{z_{1b}, \ldots, z_{rb}},$$

where Z has row margins y and column margins C'. The exact largest complete branch weight below x is

$$B(x) = A(x)L(R - x; C'). \tag{1}$$

Every value used below is positive and finite because its daughter and suffix margins are explicitly required to be feasible.

There are two numerical scopes. The research oracle uses arbitrary-precision integer path weights, exact tail membership, exact rational output and replayable certificates. The native product uses floating log-factorials, the FEXACT comparison convention with tolerance 3.45254×10^{-7}, and floating output. Release tests find the same conditional tail on their declared cases, but the scopes do not provide identical numerical guarantees. Certificate generation and verification are timed separately from uncertified native computation.

3 Discrete-convex suffix structure

The feasible suffix-margin domain is

$$\mathcal{D}(C') = \left\{ y \in \mathbb{Z}_+^r : \sum_a y_a = \sum_b c_b \right\}.$$

Because the transportation graph is complete bipartite, this is exactly the domain of feasible row margins for C', including vectors with zero coordinates.

The M-concavity convention in this paper is the following exchange axiom. For a finite function f on a fixed-sum integral domain, if $u_a > v_a$, there is a coordinate b with $u_b < v_b$ such that

$$f(u) + f(v) \le f(u - e_a + e_b) + f(v + e_a - e_b). \tag{EXC}$$

The exchanged vectors must belong to the domain. This is equation (2.4) on printed page 277 of Murota [5].

Lemma 1 (Network-induced suffix M-concavity)**.** *The function $\ell(y) = \log L(y; C')$ is M-concave on $\mathcal{D}(C')$.*

Proof. For a suffix table $Z = (z_{ab})$,

$$\log L(y; C') = \sum_b \log(c_b!) - \min_Z \sum_{a,b} \log(z_{ab}!). \tag{2}$$

Thus ℓ is a constant minus a separable convex transportation minimum. More explicitly, map z_{ab} to integer flow on the complete bipartite network from row vertices to column vertices. Its row boundary is y, its column demand is C', and each arc has cost

$$\phi(n) = \log(n!), \qquad \phi(n+1) - \phi(n) = \log(n+1).$$

The increments are nondecreasing, so ϕ is discrete convex. The negative optimum value of this separable convex-cost flow is M-concave in its boundary. This is the min-cost-flow instance in Murota's Example 2.3, printed page 279, and follows from his network induction theorem, Theorem 7.2, printed page 305 [5]. The same sign mapping appears in convex form in Murota's Theorem 4.7(2), printed pages 26–27 [6]. The first term of (2) is independent of y, proving the claim. □

Zero residual coordinates are permitted by $\mathcal{D}(C')$. An exchange decrements only a strict surplus and cannot create a negative coordinate. If C' is empty, then $\mathcal{D}(C') = \{0\}$, $L(0; \varnothing) = 1$, and the exchange statement is vacuous. Neighboring vectors outside the domain are not assigned approximate or surrogate values; statements below require all displayed neighbors to be feasible.

4 Strict within-line order

The following standard balancing argument makes the mode argument self-contained. One-dimensional daughter ordering and its discrete-convex precedents include Requena and Martín Ciudad [7].

Lemma 2 (Balanced integer split)**.** *Let Z be a nonnegative integer matrix with even row and column sums, and suppose specified entries of Z are even. Then $Z = U + V$ for nonnegative integer matrices U, V that have half the row and column sums of Z, satisfy $|u_{ab} - v_{ab}| \le 1$ in every cell, and split every specified even entry equally.*

Proof. Split every even entry equally. Make a bipartite graph with one edge for each odd entry, joining its row and column vertices. Each vertex has even degree because its margin and the contributions of even entries are even. Decompose the graph into cycles and color each cycle alternately. Give the larger half of each odd entry to U on one color and to V on the other. The two colors occur equally often at each vertex, which gives the required margins and cellwise balance. □

Lemma 3 (Strict within-line log-concavity)**.** *Choose distinct rows p, q, set $x(t) = x(0) + t(e_p - e_q)$, and let $I \subset \mathbb{Z}$ be the feasible integer interval. For every adjacent triple in I,*

$$B(x(t))^2 > B(x(t-1))B(x(t+1)). \tag{3}$$

Consequently the adjacent ratios $B(x(t+1))/B(x(t))$ strictly decrease in t. The line has either one maximum or two adjacent equal maxima. Its leftmost maximizer is

$$p = \min\left(\{t \in I : t < \max I,\ B(x(t+1))/B(x(t)) \le 1\} \cup \{\max I\}\right). \tag{4}$$

Proof. Let Y^- and Y^+ minimize the products of cell factorials among complete tables whose first columns are $x(t-1)$ and $x(t+1)$. The matrix $Z = Y^- + Y^+$ has even margins and first column $2x(t)$. Apply Lemma 2, specifying the first column, to obtain complete tables U, V with first column $x(t)$.

For two nonnegative integers of fixed sum, their factorial product is minimized when they are as equal as possible: if $a \ge b + 2$, moving one unit from a to b multiplies $a!b!$ by $(b+1)/a < 1$. Applying this cellwise gives

$$D(U)D(V) \le D(Y^-)D(Y^+), \qquad D(W) = \prod_{a,b} w_{ab}!.$$

The inequality is strict in first-column row p, whose outer values are $x_p(t) - 1$ and $x_p(t) + 1$, while both middle values are $x_p(t)$. Hence, writing $d(x)$ for the minimum of D at first column x,

$$d(x(t))^2 \le D(U)D(V) < d(x(t-1))d(x(t+1)).$$

Taking reciprocals and multiplying by the common squared product of column factorials proves (3).

Dividing (3) by adjacent positive terms shows that the forward ratios strictly decrease. They can therefore cross one at most once. A ratio equal to one produces exactly two adjacent equal maxima; otherwise the maximum is unique. Before the first ratio at most one, values strictly increase, and at and after it they do not increase. This proves (4), including intervals of length one. □

5 Strict cross-slice rhombus

Choose distinct rows i, j, k. Suppose all four daughters

$$x, \quad x^i = x + e_i - e_j, \quad x^k = x + e_k - e_j, \quad x^{ik} = x + e_i + e_k - 2e_j$$

are feasible. In particular, feasibility of x^{ik} requires $x_j \ge 2$.

Theorem 1 (Strict cross-slice rhombus)**.**

$$B(x^i)B(x^k) > B(x)B(x^{ik}). \tag{5}$$

Proof. Set $y = R - x$ and $y^{ik} = R - x^{ik}$. Then

$$y - y^{ik} = e_i + e_k - 2e_j. \tag{6}$$

Relative to y^{ik}, the vector y has surplus coordinates i and k, and j is the only coordinate with the required opposite sign. After selecting surplus coordinate i in (EXC), the exchange recipient is therefore forced to be j.

The four suffix margins, written explicitly, are

daughter	suffix margin
x	y
x^i	$y - e_i + e_j$
x^k	$y - e_k + e_j$
x^{ik}	$y - e_i - e_k + 2e_j$.

Lemma 1 instantiates (EXC) as

$$\ell(y) + \ell(y - e_i - e_k + 2e_j) \le \ell(y - e_i + e_j) + \ell(y - e_k + e_j).$$

Exponentiating gives the known, non-strict suffix inequality

$$L(y - e_i + e_j; C')L(y - e_k + e_j; C') \ge L(y; C')L(y - e_i - e_k + 2e_j; C'). \tag{7}$$

Direct cancellation of the four first-column multinomials gives

$$\frac{A(x^i)A(x^k)}{A(x)A(x^{ik})} = \frac{x_j}{x_j - 1}. \tag{8}$$

Feasibility of x^{ik} implies $x_j \ge 2$, so the factor is strictly greater than one. Multiplying (7) and (8) proves (5). □

The conceptual separation is exact: known network-induced M-concavity gives the weak inequality; the first-column multinomial gives the strict factor; their product gives the strict branch rhombus.

6 Monotone leftmost modes

Fix all daughter coordinates outside rows i, j, k. Let d be the allocation total remaining in these three coordinates and write

$$t = x_i, \qquad z = x_k, \qquad x_j = d - z - t.$$

Let r_i, r_j, r_k be the three residual row capacities. For an integer z with $0 \le z \le r_k$, define

$$I_z = [\ell_z, u_z] \cap \mathbb{Z}, \qquad \ell_z = \max(0, d - z - r_j), \qquad u_z = \min(r_i, d - z). \tag{9}$$

The slice is feasible exactly when $I_z \ne \varnothing$. For adjacent feasible slices $z, z+1$,

$$\ell_{z+1} = \max(0, d - z - 1 - r_j) \le \ell_z, \qquad u_{z+1} = \min(r_i, d - z - 1) \le u_z. \tag{10}$$

Lemma 4 (Monotone leftmost mode)**.** *Let $b(t, z) > 0$ be defined on feasible integer intervals $I_z = [\ell_z, u_z] \cap \mathbb{Z}$. Assume that:*

1. *$b(t, z)^2 > b(t-1, z)b(t+1, z)$ for every feasible adjacent triple;*
2. *the forward ratio $\rho_z(t) = b(t+1, z)/b(t, z)$ satisfies $\rho_{z+1}(t) < \rho_z(t)$ wherever both ratios exist;*
3. *$\ell_{z+1} \le \ell_z$ and $u_{z+1} \le u_z$.*

Then the leftmost maximizer $p(z)$ weakly decreases:

$$p(z+1) \le p(z).$$

Proof. Strict log-concavity makes $\rho_z(t)$ strictly decreasing in t. Thus Lemma 3's characterization (4) applies to each slice, including a slice of length one and the case of two adjacent equal maxima.

Write $p = p(z)$ and $I_{z+1} = [\ell', u'] \cap \mathbb{Z}$. If $p > u'$, then $p(z+1) \le u' < p$; this is the upper-endpoint clipping case. Otherwise, $\ell' \le \ell_z \le p \le u'$, so the previous mode remains feasible. If $p = u'$, the desired inequality is immediate. If $p < u'$, the ratio $\rho_{z+1}(p)$ exists. Because $u' \le u_z$, the ratio $\rho_z(p)$ also exists. The leftmost-mode characterization gives $\rho_z(p) \le 1$, with equality precisely when p is the left member of two adjacent equal maxima. Assumption 2 then gives

$$\rho_{z+1}(p) < \rho_z(p) \le 1.$$

Therefore the first ratio at most one on slice $z+1$, or its upper endpoint if no such ratio exists, occurs no later than p. Hence $p(z+1) \le p$. This also covers ratios at the interval endpoints: a forward ratio is used only when its right neighbor exists. □

Corollary 1 (Fisher-network slice order)**.** *For $b(t,z) = B(x)$ under (9), the leftmost maximizer weakly decreases with z.*

Proof. Lemma 3 gives assumption 1 of Lemma 4, and (10) gives assumption 3. Theorem 1, with increments in t and z, is

$$B(t+1,z)B(t,z+1) > B(t,z)B(t+1,z+1).$$

Dividing by positive terms yields

$$\rho_{z+1}(t) = \frac{B(t+1,z+1)}{B(t,z+1)} < \frac{B(t+1,z)}{B(t,z)} = \rho_z(t),$$

which is assumption 2. Apply Lemma 4. □

Lemma 5 (Amortized peak-query bound)**.** *Consider s consecutive feasible slices of width at most $m+1$. Find the first leftmost mode by binary search. On every later slice, start at the preceding mode clipped to the new upper endpoint and walk left to the new leftmost mode. With branch values cached within the sweep, peak location uses*

$$O(\log m) + O(m) + O(s) = O(m+s)$$

branch-bound queries and peak comparisons. Independent binary searches use $O(s \log m)$.

Proof. Write p_h for the leftmost mode of slice h, and for $h \ge 1$ let $q_h = \min(p_{h-1}, u_h)$ be the clipped starting point. The corollary and feasibility give

$$p_h \le q_h \le p_{h-1}.$$

The number of left moves on slice h is $q_h - p_h$, and therefore

$$\sum_{h=1}^{s-1}(q_h - p_h) \le \sum_{h=1}^{s-1}(p_{h-1} - p_h) = p_0 - p_{s-1} \le m.$$

There is one terminal check per later slice, and each ratio comparison requests only a constant number of cached branch values. The first binary search costs $O(\log m)$. This proves the displayed bound, with the cases $m \le 1$ absorbed into the constant. □

Lemma 5 covers branch-bound calls and peak-location comparisons. Daughter generation, threshold searches, accepted intervals, forwarded arcs, residual states and total runtime depend on separate mechanisms. Peak comparisons can therefore fall without a corresponding wall-clock improvement.

7 Algorithmic consequence

The shared native C core exposes five explicit modes: `baseline`, `accept_only`, `line_batch`, `line_batch_monge` and `auto`. The two line modes have identical daughter generation, exact outer-interval membership, interval-mass aggregation, reduced forwarding and prefix-state reuse. They differ only in peak location: `line_batch` searches each slice independently, whereas `line_batch_monge` uses Corollary 1. This is the theorem-only ablation.

The native engine supports $3 \times c$ tables and, by safe transposition, $r \times 3$ tables. Other dimensions are explicitly delegated by the R interface to `stats::fisher.test(hybrid=FALSE, simulate.p.value=FALSE)`; the standalone program reports them as unsupported. The product is not claimed as a general $r \times c$ replacement.

Accept-only pruning is an exact engineering mode. It omits shortest-path rejection and retains longest-bound acceptance; every unresolved daughter is forwarded. It preserves tail membership but can expand excessively. The current `auto` rule selects `line_batch`; no stable adaptive selection rule was found.

8 Implementation and controlled ablations

The native interface exposes `baseline`, `accept_only`, `line_batch`, `line_batch_monge`, and `auto`. The current auto rule selects `line_batch`; no stable adaptive selection rule was found.

All quoted measurements below come from the frozen machine-readable product benchmark. Medians were measured on Apple arm64 with R 4.6.1. Native-core timing repeats the shared C call in one process; R timing excludes R startup; certificate work is excluded.

Family	line core	R + line	base R	pinned FEXACT
small threshold tie	0.007 ms	0.038 ms	0.500 ms	0.305 ms
repeated prefixes	0.288 ms	0.700 ms	1.000 ms	0.541 ms
wide first column	0.256 ms	0.600 ms	1.000 ms	0.414 ms
retained 10 ms tier	0.476 s	0.780 s	0.054 s	0.0619 s
retained 100 ms tier	1.817 s	2.830 s	0.198 s	0.1988 s
retained 1 s tier	workspace	workspace	2.322 s	2.447 s
retained 10 s tier	workspace	workspace	10.367 s	11.424 s

The package can beat `stats::fisher.test` on small supported tables where setup costs dominate. It is about 8–15 times slower on the completed 10 ms and 100 ms stress tiers. At the 10 ms tier, `line_batch` takes 0.476 s versus 1.640 s for `accept_only`; internal peak allocation is 171 MB versus 380 MB. Line batching forwards 9.86 million daughters and accepts 19.69 million line daughters without forwarding. At the 100 ms tier, line batching completes with 310 MB internal peak allocation while the unbatched modes exceed the configured 512 MiB workspace. All package modes exceed that workspace on the retained 1 s and 10 s tiers. These are batching, interval-aggregation, reduced-forwarding and prefix-reuse effects, not consequences attributed to the rhombus.

Theorem-only family	bound calls saved	comparisons saved	Monge/independent time
wide first column	308	625	0.949
repeated prefixes	87	353	1.010
retained 10 ms tier	4,175	16,104	0.995
retained 100 ms tier	11,439	32,609	0.986

Residual states, considered and forwarded arcs, accepted intervals, line searches, prefix work, peak movement and internal peak allocation are identical within each theorem-only pair. Query savings are consistent, but timing ratios are small and change sign. The supported conclusion is query reduction without a stable theorem-only runtime or memory benefit.

The release correctness gate compares the exact-rational oracle, all manual native modes, `stats::fisher.test`, and representatives through the pinned R 4.6.1 FEXACT harness. It covers 1,198 cross-language cases, including exhaustive small tables, deterministic random tables, permutations, transposes, ties, sparse and dense tables, zero cells, large path weights, workspace and allocation failures, and deterministic interruption.

9 Empirical boundary and failed conjectures

Fixed-threshold boundaries need not move monotonically and need not move one unit at a time; retained exact counterexamples begin at total eight. A separate shared-longest-DP conjecture also failed: a root call already closes the reachable suffix DAG, so deleting later top-level calls need not delete internal dynamic-programming states.

The package remains experimental, three-row only, and less memory-efficient than FEXACT on hard workloads. Integrating the cross-slice walk into R's FEXACT core would require a larger architectural change: FEXACT canoni-

calizes residual rows and emits daughters individually, whereas the walk needs stable row labels and ordered adjacent slices.

10 Attribution and scope of contribution

This work builds on the cited exact-testing, network-algorithm and discrete-convexity literature. In particular, the Mehta–Patel network, Algorithm 643/FEXACT and its refinements, network-induced M-concavity, transportation formulations, path trimming and established one-dimensional daughter ordering remain the contributions of their respective authors [3, 4, 2, 5, 6, 1, 8, 7]. This paper contributes the strict cross-slice statement and its bounded algorithmic consequence, together with an independently written research implementation and controlled performance evidence. Citations identify intellectual provenance; they do not imply endorsement by the cited authors or institutions.

11 Reproducibility and computational assistance

The public implementation corresponding to this paper is available at

> https://github.com/Maar10Herr/Exact-Fisher-Network-Public.

It contains the shared C core, R and command-line interfaces, the exact-rational Python reference, and deterministic verification tests. The public checks are run with `make test`; the R package is built and checked separately with `make check-r`.

The frozen product benchmark underlying Section 8 is identified by

> SHA-256:
> `47e177675c61cbbf5d095d1531678276785beadbd4c1c88bc0dac2e04f335f90`.

AI tools assisted with literature organization, conjecture exploration, drafting, implementation and testing. Sources were checked directly; mathematical claims reduce to the cited network-induction result and explicit factorial arguments, and computational claims are checked against exact oracles, differential tests, controlled counterexamples and frozen compiled benchmarks.

12 Conclusion

Known suffix M-concavity and a strict multinomial factor give a strict cross-slice rhombus for exact Fisher-network branch values. With a self-contained within-line lemma and explicit moving-domain argument, the rhombus orders leftmost modes across slices and reduces exact peak queries. The controlled native ablation finds this theorem-level change timing-neutral. Line batching improves matched unbatched modes on prefix-heavy tables, while base R FEXACT remains substantially faster on retained difficult workloads. The theorem, implementation and benchmark together define a reproducible foundation for future exact-network work.

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